UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	))	Docket No. RM26-4-000

REPLY COMMENTS
OF
FIRSTENERGY SERVICE COMPANY

Pursuant to the Federal Energy Regulatory Commission's ("Commission") October 27, 2025, Notice Inviting Comments on the Department of Energy's proposed advance notice of proposed rulemaking ("ANOPR") requested under section 403 of the Department of Energy Reorganization Act[1] on October 23, 2025, and the Commission's Notice Granting Extension of Time on November 7, 2025, FirstEnergy Service Company ("FirstEnergy")[2] respectfully submits the following reply comments.[3]

EXECUTIVE SUMMARY

In these reply comments, FirstEnergy responds to comments submitted on November 21 in order to reinforce the approach that it believes will best achieve the objectives of the Department of Energy's (the "Department"), September 18, 2025, Speed to Power Initiative.[4] As the initial

[1] 42 U.S.C. § 7173.

[2] FirstEnergy also submits these comments on behalf of its transmission-owner affiliates American Transmission Systems, Incorporated, Jersey Central Power & Light Company, Mid-Atlantic Interstate Transmission LLC, The Potomac Edison Company, Monongahela Power Company, Keystone Appalachian Transmission Company, and Trans-Allegheny Interstate Line Company.

[3] FirstEnergy is also joining reply comments submitted jointly with certain other PJM Interconnection, L.L.C. ("PJM") transmission owners. FirstEnergy encourages the Commission to read these reply comments in conjunction with those.

[4] Department of Energy, *Energy Department Launches Speed to Power Initiative, Accelerating Large-Scale Grid Infrastructure Projects* (Sept. 18, 2025), https://www.energy.gov/articles/energy-department-launches-speed-power-initiative-accelerating-large-scale-grid.

comments taken together all demonstrate, any federal rules governing the interconnection of large load customers should be narrowly tailored to achieving the Administration's speed to power mandate as quickly, affordably, and efficiently as possible.

- First, the Commission should not require that all network upgrades that are determined to be necessary solutions to interconnecting large load customers be included as projects subject to competition in regional transmission planning processes.
- Second, the Commission should not require transmission providers to publicly disclose customer specific load profiles or submit annual reports containing customer cost impacts because customer specific information is confidential.
- Third, the Commission should not *require*, but rather, could encourage, where feasible, use of artificial intelligence ("AI") or automation tools in the interconnection study process.
- Fourth, the Commission should not *require*, but may encourage, analysis of the efficacy of network upgrade solutions that include grid enhancing technology or advanced transmission technology components.
- Fifth, netting large load with generation is inconsistent with reliability standards; therefore, the Commission should not direct transmission providers to use such a practice.
- Sixth, system protection facilities are not sufficient to ensuring reliability of the system in the case of netting large load with generation.
- Seventh, to the extent the Commission elects to work toward a final rule, it should define a "large load customer" to mean a data center customer whose facility is at least 200 MW.
- Finally, the Commission should continue to rely on its long-standing policy of rolling the costs of integrated network upgrades into network transmission rates.

Awareness of each of these measures is paramount to helping achieve the Department's speed to power goals.

I. REPLY COMMENTS

A. The Commission Should not Require That Transmission Providers Apply Regional Planning and Competition Rules to Large Load Interconnections.

LS Power Development, LLC ("LS Power") argues that large load integration should be managed within the same coordinated regional planning and study framework used for generation and reliability and that the Commission should encourage competition for the construction of large load-related infrastructure.[5] Issues regarding the appropriate scope of regional versus local planning and competition have been comprehensively addressed and decided by the Commission in those prior orders. There is no reason to re-open those issues now, and certainly not in this proceeding. LS Power's requests will defeat the Department's "speed-to-power" goal. Introducing these contentious issues into this rulemaking would substantially complicate the proceeding and jeopardize the Commission's ability to act quickly. Moreover, there is now significant evidence that Order No. 1000 competitive transmission processes are not having the intended result in terms of either efficiency or cost-effectiveness. Also, given the magnitude of infrastructure upgrades that could be driven by data center load additions over the next several years, such a requirement would likely add significant time if all such upgrades are subject to what is often a lengthy competitive solicitation process. In any event, large load interconnections that require regional transmission upgrades already are considered as part of PJM's Regional Transmission Expansion Plan process.

B. Additional Public Posting Requirements are Neither Appropriate nor Necessary.

The Office of the Ohio Consumers' Counsel ("OCC") requests that the Commission require public utilities and Regional Transmission Organizations ("RTOs") to publicly post: (1) all interconnection requests 20 MW and above; (2) cost responsibility determinations and any resulting upgrades; and (3) annual reporting on consumer cost impacts of large load interconnections, associated transmission expansions, and impact on capacity market rates.[6] This request, however, is beyond the scope of the ANOPR and if implemented will slow any progress toward achieving the Department's "speed to power" goal. FirstEnergy is a strong proponent of

[5] LS Power at 8, 13.

[6] OCC at 3-4.

transparency when that transparency benefits customers. However, customer-specific information is often confidential under state law or contract, in which case FirstEnergy is required to maintain such information as strictly confidential unless the customer otherwise consents to its disclosure. Also, a good deal of the transmission information that OCC requests (*e.g.*, costs of transmission upgrades) is already provided through publicly-available sources. For example, information regarding project status and cost estimates for transmission upgrades is already posted publicly on PJM's website.[7] There is no good reason to expand the scope of the ANOPR to create yet another forum to address and re-litigate the amount of transmission planning and cost information provided to stakeholders.[8]

C. The Commission Should not Require the use of Automation and AI Tools.

The Data Center Coalition ("DCC") argues that the Commission should require that RTOs and other transmission providers adopt automation and AI tools to align study timelines with needed urgency.[9] However, AI tools at present are not sufficiently advanced to meaningfully accelerate planning and load analyses, although FirstEnergy anticipates that AI tools will eventually play an important role in interconnection studies. Also, the Commission should avoid regulating the study practices of transmission providers at the detailed level of dictating the use of particular tools or programs. Such rules would be difficult to define and enforce, and prohibiting flexibility for transmission providers to use the right tool for each specific job could end up

[7] PJM, *Project Status & Cost Allocation*, https://www.pjm.com/planning/m/project-construction (last visited Dec. 5, 2025).

[8] Granting OCC's request could also raise jurisdictional concerns. Large load interconnections invariably involve requests for retail service, which are governed by State rules, including rules governing the type and amount of information that interconnecting utilities can make public. The Commission should avoid potential conflicts with state-jurisdictional retail service provisions, and the resulting litigation and uncertainty that such conflicts will entail.

[9] DCC at 11.

undermining rather than enhancing efficiency, thereby working contrary to the ANOPR's speed-to-power goal.

D. The Commission Should not Mandate the Evaluation or use of GETs and ATTs.

A few parties suggest that the Commission should require that grid-enhancing technologies ("GETs") and advanced transmission technologies ("ATTs") be evaluated in building infrastructure associated with large load interconnections, or encourage their use.[10] As with automation and AI tools, transmission providers already look for opportunities to use GETs and ATTs. However, such technologies are not always useful. Nor are they appropriate substitutes for constructing needed transmission capacity.[11] Thus, although transmission providers should consider GETs and ATTs in connection with large load interconnections, the Commission should not mandate the evaluation and use of GETs and ATTs. Layering such specific requirements on transmission providers will increase the burdens and complexity of compliance with any final rule and thus could frustrate the Department's "speed to power" goal.

E. Netting Load With Generation is Inconsistent With NERC Standards.

Studying the net impact of hybrid facilities undermines the North American Electric Reliability Corporation ("NERC") transmission planning performance standards, which require testing the ability of the transmission system to withstand specific contingencies including loss of load and loss of generation. Certain commenters attempt to highlight the benefits of the netting

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[10] *See* Industrial Energy Consumers of America, the American Forest & Paper Association, the PJM Industrial Customer Coalition, and the Coalition of MISO Transmission Customers, at 27-28 ("Industrial Customer Organizations"); *see also* Clean Air Task Force, Inc., Coalition for Affordable Utility Services and Energy Efficiency in Pennsylvania; Earthjustice; Environmental Defense Fund; Montana Environmental Information Center; Natural Resources Defense Council, Sustainable FERC Project; Sierra Club; and Southern Environmental Law Center, at 7-8 ("PIOs").

[11] Their utility and purpose also are remote from the primary goals of the proposed ANOPR, putting them beyond the scope of the ANOPR.

principle, but fall short of articulating anything that would justify the Commission directing the industry to ignore their responsibilities as transmission providers and reliability coordinators.[12] Netting fails to provide the redundant protection necessary for the system to perform reliably. Under the current PJM Open Access Transmission Tariff, and the tariffs of most other Independent System Operators ("ISOs") and RTOs, the interconnection of load and generation are studied separately in accordance with NERC's standards. The reason for this bifurcation is to verify that the potential impact of the generator on the transmission system is comprehensively evaluated, and that any large load is able to be reliably served. For instance, if the transmission network is only planned to support the net withdrawal and export at the point of interconnection, the interconnection capability may be undersized compared to the amount of generation that could be made available at a time when the load customer is curtailed or not consuming power.

Commenters supporting study of only the net amount[13] fail adequately to account for equipment failure or NERC standards to assure reliability in the event of failure. Were the Commission to adopt this netting approach, the resulting reliability deficiencies would frustrate the Department's desire to accelerate speed-to-power for large load customers. The New York State Department of State Utility Intervention Unit correctly explains that, a study resting on a momentary snapshot of a highly variable process is a recipe for reliability issues down the road, and "impose[s] unnecessary risks to the bulk power system."[14] Also, if an RTO or ISO "does not

[12] Vantage Data Centers at 6-7; PIOs at 24; Pennsylvania Office of Consumer Advocate and Delaware Division of the Public Advocate at 20 ("Pa. & Del. Advocates").

[13] *See* OpenAI Inc. at 2; *see also* Vantage Data Centers at 6-7.

[14] The New York State Department of State Utility Intervention Unit at 6.

study the impact of the entirety of the load," it simply does not know what will happen when generation does not provide its full capacity, as Southwest Power Pool Inc. ("SPP") observes.[15]

Finally, netting raises another issue not addressed in the initial comments: it undermines NERC standards and therefore also clashes with the ANOPR's principle fourteen.[16] NERC Standard TPL-001 establishes Transmission System Planning Performance Requirements, and is designed to model the reliability of the BES under various contingencies. Transmission planners must study, among other events, loss of load and loss of generation, and do so separately.[17] That requirement simply drives home the fact that hybrid loads are variable, and that real-world contingencies do not add up load and generation. More importantly, it makes little sense to net load and generation at the initial study stage while demanding separate reliability measurements later on.

FirstEnergy implements and supports NERC's transmission planning standards and concludes that the system is most reliable when those standards are observed; therefore, netting should not be permitted under the current reliability framework.[18]

F. System Protection Facilities Cannot Alone Ensure Against System Failure.

Several commenters suggest or at least concede that system protection requirements for hybrid facilities are a good idea.[19] A seat belt is also advisable when traveling at a high rate of speed, but if the seat belt fails, an air bag will help mitigate the impact. The employment of

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[15] SPP at 11.

[16] Transmission System Planning Performance Requirements, NERC Reliability Standard TPL-001-5.1; ANOPR at P 30.

[17] *See* TPL-001-5.1 at 26 (Table 1).

[18] Netting cannot be reasonable considered until NERC completes its development of reliability guidelines for large load interconnection requirements on or about June 2026.

[19] Meta Platforms, Inc. at 6 ("Meta"); Tri-State Generation and Transmission Association, Inc. at 7; LS Power at 11; PIOs at 25.

transmission system redundancies are necessary to meet the Department's speed to power mandate. System protection facilities alone cannot ensure necessary reliability, which in turn threatens the ability to safely and expeditiously interconnect large loads.

Protective equipment that would enforce a net amount is not a failsafe; to the contrary, such equipment suffers from an approximately 6% failure rate.[20] That observation points to netting's primary hurdle: "what may be hundreds of MWs of load variability."[21] This is true even with protection facilities designed to prevent unauthorized injections or withdrawals, because "contingencies and control failures routinely defeat netting."[22] The point of an interconnection study is figuring out how to deal with unexpected contingencies—and 6% is not even so unexpected. System failures in this context are even more dangerous given the abnormal size of large loads. The Maryland Public Service Commission highlighted a recent simultaneous loss of about 1,500 MW of data center load in Northern Virginia.[23] Only skill and good fortune averted serious consequences, but that is no substitute for proper study and planning. And that outage did not even involve hybrid facilities, where levels of both load and generation are variable.

Besides, *contra* Vantage, netting misses real-world grid impacts in another way. A hybrid load's real-world impact on the grid is not a number: it is a set of simultaneous injection and

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[20] NERC, *M-9 Protection System Misoperations*, https://www.nerc.com/programs/reliability-assessment--performance-analysis/reliability-indicators/m-9-protection-system-misoperations (last visited Dec. 1, 2025). The misoperation rate is the number of protection system misoperations divided by the total number of protection system operations. *See* NERC, *PAS Metric, M-9 Protection Systems Misoperation Rate*, https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/pas/m-9_correct_protection_system_operations.pdf (last visited Dec. 1, 2025). *See also* Reliability First, *2025 Misoperation Assessment*, Sept. 2025, https://www.rfirst.org/wp-content/uploads/2025/10/2025-RF-Misoperation-Assessment.pdf.

[21] New York Transmission Owners ("NYTOs") at 19. Variability also creates difficulty applying tariffs, the Transmission Owners explained, a comment the Midcontinent Independent System Operator, Inc. ("MISO") echoed in noting that netting "may require significant tariff and BPM revisions." MISO at 24.

[22] Large Public Power Council ("LPPC") at 13.

[23] Maryland Public Service Commission at 8-9.

withdrawal processes, each with its own unique effects. The Commission's hypothetical facility does not stop generating 600 MW and withdrawing 500 MW—with all attendant impacts—simply because the sum of those numbers equals a 100 MW injection.[24] Those impacts may include adverse interactions between control systems, a problem netting-based studies would miss.[25] Studying the generation and load separately therefore remains, as MISO noted, "critical for reliability purposes,"[26] and equipment with a 6% failure rate is no substitute.

Meanwhile, NERC standard PRC-012-2 requires transmission planners to evaluate both the failure and inadvertent operation of system protection facilities—precisely the kind of contingency netting (by assuming perfectly-functioning protection systems) obscures.[27] Here, too, it would be cheaper and more efficient in the long term to discover potential issues up front, rather than patching up reliability problems after interconnection, with the attendant curtailments and outages that would likely occur.

In short, system protection facilities are insufficient to control and mitigate large load withdrawals at a netted point of interconnection. Other system redundancies are therefore required. The failure of system protection facilities must be considered so the impact of the failure can be addressed to ensure the reliability of the transmission system. FirstEnergy does not support large load interconnection procedures that seek to justify the netting of load and generation at the

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[24] *See* Entergy Services, LLC at 27.

[25] LPPC at 13-14.

[26] MISO at 24.

[27] Remedial Action Schemes, NERC Reliability Standard PRC-012-2 R4. A Remedial Action Scheme is a "scheme designed to detect predetermined System conditions and automatically take corrective actions that may include, but are not limited to, adjusting or tripping generation (MW and Mvar), tripping load, or reconfiguring a System(s)." It is designed to maintain BES stability, and includes the kinds of system protection facilities large loads would use to prevent unauthorized injections or withdrawals. *See* NERC, *Glossary of Terms Used in NERC Reliability Standards*, Oct. 1, 2025, https://www.nerc.com/globalassets/standards/reliability-standards/glossary_of_terms.pdf.

same point of interconnection so long as such interconnection is supported by system *protection* facilities. These purported measures are insufficient to address the reliability deficiency and fail under current NERC standards. For instance, Remedial Action Schemes are temporary solutions and do not provide transmission planners with the necessary confidence in designing a resilient and reliable system built for exponential load growth. NERC requires planners to reevaluate Remedial Action Schemes every five years. Interconnections must contain minimum requirements that are constructed to ensure reliability for the life of the facility.

G. The Commission Should Define a Large Load Customer as Having a Load of 200 MW or Higher.

FirstEnergy's initial comments explained that the Commission should increase the ANOPR's tentatively proposed threshold of 20 MW to at least 200 MW because doing so will prevent the final rule from frustrating the goals of the ANOPR by adversely affecting industrial, manufacturing and other large load customers who are not the intended targets of the rule, which in turn would undermine community growth, hamper efforts to reshore manufacturing, and stall or even reverse job growth.[28]

A substantial majority of commenters, representing a wide spectrum of interests, agree that the ANOPR's tentatively proposed threshold of 20 MW is too low.[29] As the Industrial Customer Organizations explain, for example, "a 20 MW limit is far too low and would capture a significant number of manufacturing entities and other industrial and institutional organizations that do not have the speed-to-market and scale demands that are primarily driving today's grid challenges."[30]

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[28] FirstEnergy at 2-4.

[29] *See, e.g.*, Organization of MISO States, Inc. at 12-13; Meta at 4; Industrial Customer Organizations at 25-27; California Public Utilities Commission at 14 ("CPUC"); Louisiana Public Service Commission & Mississippi Public Service Commission at 8; NYTOs at 12-14.

[30] Industrial Customer Organizations at 26-27.

Such a low threshold will have an adverse effect on the Commerce Department's efforts toward reshoring American manufacturing. The threshold would similarly provide the wrong economic signal to businesses looking to expand their commercial footprint in communities across the country. These effects will harm job growth during a time when the country is focused on economic growth. Diversification of the industrial base is critical to the regions in FirstEnergy's footprint. A low megawatt threshold for jurisdictional load interconnections may unintentionally propagate economic risk and instability in communities searching for economic certainty and stability. A more targeted threshold (200 MW) that avoids these impacts will not compromise the Department's speed to power mandate because the primary drivers of that goal are, and will continue to be, larger scale data centers.

To be sure, a minority of commenters support a 20 MW threshold.[31] But those comments misperceive the size of data centers and/or fail to confront the negative effects of sweeping too broadly in a rule whose goal is to facilitate prompt and equitable interconnection of data center customers. For example, OCC incorrectly asserts that a 20 MW threshold "limits federal jurisdiction to only the largest transmission-connected loads."[32] As numerous commenters explained, typical data center loads are substantially higher than 20 MW.[33] Accordingly, a 20 MW (or similarly low) threshold will mean that many more loads are in the queue for study and

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[31] *See, e.g.*, Office of the Federal Energy Advocate of the Public Utilities Commission of Ohio at 3, Pa. & Del. Advocates at 17; California Independent System Operator Corporation at 5-6 ("CAISO"); OCC at 8.

[32] *See* OCC at 8; *see also, e.g.*, Pa. & Del. Advocates at 17 (not addressing the size of typical data centers); CAISO at 5-6 (same).

[33] *See, e.g.*, Indicated PJM Transmission Owners at 4 & n.8 ("new hyperscale data centers typically possess power capacities of 100 to 1,000 MW each") (citing EPRI, *Powering Intelligence: Analyzing Intelligence and Data Center Energy Consumption* (May 28, 2024), https://www.epri.com/research/products/3002028905 at 2).

interconnection. That will delay the interconnection of the data center loads that are the ANOPR's focus.

Instead, the Commission should use at least 200 MW as the threshold and should further limit the class of customers to data center customers.

H. The Commission Should Adhere to its Long-Standing Approach of Using Rolled-In Rate Treatment for Network Upgrade Costs.

FirstEnergy's initial comments explained that, if the Commission were to impose a requirement that 100% of network upgrade costs appear on a particular retail customer's[34] electricity bill, that would arguably improperly usurp the states' jurisdiction over retail sales, and resolving this jurisdictional dispute through the appellate courts will likely delay the interconnection of data center customers. Moreover, mandating direct assignment will likely lead to the same inefficiencies that have plagued the generator interconnection process, such as numerous withdrawal and restudy cycles that often occur when individual customers trigger the need for significant network upgrades. These types of outcomes will frustrate the Department's speed-to-power goal.[35] The best way to avoid these pitfalls is to use the Commission's long-standing approach that network upgrade costs are rolled in to the transmission owner's rate base and recovered from all *wholesale* transmission customers proportionally to their load—a mechanism that automatically, over time, ensures that each wholesale customer pays its fair share of costs that benefit the entire system.[36] At a minimum, even if the Commission directly assigns

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[34] Large load customers are retail customers.

[35] FirstEnergy at 6-8.

[36] *Id.* at 7 & nn.13-14. *See also* Industrial Customer Organizations at 9-10 ("In evaluating where to make long-term investments in communities, manufacturers have relied on [the rolled-in ratemaking] approach to cost allocation for years, as it is supported by cost causation principles."); ("[T]he ANOPR overlooks the practical and financial consequences of potentially subjecting manufacturers (who are trade-exposed, labor intensive, and price sensitive) to a mandatory large load interconnection rule in which they pay 100% of upgrade costs, even if the ensuing network up-grades benefit other customers."); ("While certain large loads

network upgrade costs triggered by interconnection of a large load customer to that customer, the Commission should clarify that the transmission owner is entitled to earn a reasonable return on the investment.

1. Numerous commenters agree with FirstEnergy that interconnection of ultimate end-use retail customers, and particularly which costs appear on such customers' bills, are arguably beyond the Commission's jurisdiction under the Federal Power Act.[37] The Commission can avoid these questions, and thereby best achieve speed-to-power objectives, by limiting its rule to directing how costs are assigned by the transmission owner to *wholesale* transmission customers, leaving it to state commissions to regulate how those wholesale transmission costs are allocated among their *retail* customers. This approach would not only further cooperative federalism, but prevent the possibility of delays in interconnection while the jurisdictional issue is resolved by the courts.

2. The Commission, acting within its unquestioned scope of authority concerning allocation of network upgrade costs among wholesale transmission customers, should use the traditional rolled-in rate framework. That framework is equitable because it assigns costs proportionally based on wholesale transmission customers' use of the network. Thus, the emergence of a large load retail customer that increases the load of its load-serving entity (the wholesale transmission customer) will result in that load-serving entity having a higher load relative to other wholesale transmission customers and being assigned more of the network upgrade costs.

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may be able to commit to pay for all network system upgrades [in a direct assignment construct], such an approach would be financially devastating to manufacturers and discourage manufacturers from onshoring, engaging in redevelopment, and expanding new facilities or lines of production.").

[37] *See, e.g.*, Industrial Customer Organizations at 11-16; Attorneys General and State Consumer Advocates at 5-6; Pennsylvania Public Utility Commission at 8; CPUC at 3-11.

To be sure, that equity is achieved over time rather than in a snapshot. For example, it may be that a large network upgrade is triggered by a particular large load customer, and that upgrade enables *future* retail customers to interconnect without the need for additional network upgrades. Given these timing issues, there is possible merit, in theory, to an initial 100% assignment of network upgrade costs to the relevant wholesale transmission customer, followed by a credit back to that customer over some period of time.[38] But in practice, such a crediting approach may lead to disputes and litigation, and consequently may frustrate the paramount goal of speedy interconnection. Given this, rolled-in rate treatment remains the more tested and preferred approach.

One comment incorrectly asserts that "data from the PJM region show that transmission upgrades driven by data center loads have not led to system benefits, but rather only provide the benefit of allowing the large load to operate."[39] For example, this comment relies on the Commission's proceeding in Docket No. ER24-2172, claiming that PJM "confirmed upgrades would be triggered above 480 MW, *with no broader system benefit*."[40] In fact, neither the Commission's Order,[41] nor the PJM Transmittal Letter it cites,[42] supports the italicized language.

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[38] Applying this methodology at the wholesale level, together with other protective approaches such as the relevant wholesale transmission customer posting collateral and committing to a minimum demand factor, would protect other wholesale transmission customers—a concern that some commenters raised. *See, e.g.*, Buckeye Power, Inc. at 14-17; PIOs at 28.

[39] PIOs at 33.

[40] *Id.* (emphasis added).

[41] *PJM Interconnection, L.L.C.*, 189 FERC ¶ 61,078, at P 10 (2024) ("PJM states that it has determined that any load addition in excess of 480 MW would result in generation deliverability violations and require installation of system upgrades.").

[42] PJM Transmittal, Docket No. ER24-2172, at 4 (filed June 3, 20224) ("PJM has also determined that any load addition in excess of 480 MW would result in generation deliverability violations and require installation of system upgrades, at Susquehanna's expense, necessary to mitigate these violations in order to ensure system reliability.").

And another proceeding cited by this comment as finding no system benefit, in fact found that *there would be a system benefit*: the upgrade will "alleviate[] stability and operational constraints in the area."[43]

3. At a minimum, regardless whether and how the Commission assigns network upgrade costs to large load customers, the Commission should clarify that the transmission owner can earn a reasonable rate of return on such costs. As the Indicated PJM Transmission Owners explained,[44] doing so is required by *Hope* and *Bluefield* because transmission owners bear the risks of operating the transmission system, and is also necessary to attract investment in transmission owners that can be used to win the AI race and to respond to the national energy emergency of "an increasingly unreliable grid."[45]

I. Large Load Customers Should not be Given an Option to Build.

FirstEnergy continues to oppose the ANOPR's ninth principle, which would extend to large load customers essentially the same option to build provided to generator interconnection customers in Order No. 845. That principle clashes with the ANOPR's "speed to power" aims: it will introduce delays, inefficiencies, and reliability issues.

Numerous other transmission owners—entities experienced with Order No. 845, and therefore well-suited to comment on an option-to-build extension—have expressed similar concerns. As PPL Corporation ("PPL") explains, the generator option to build has not worked well, requiring utilities to oversee and integrate low-quality or even dangerous customer-built

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[43] PJM, *Final Reliability Analysis Report 2024 RTEP Window 1* at 56, (Feb. 10, 2025), https://www.pjm.com/-/media/DotCom/committees-groups/committees/teac/2025/20250304/20250304-2024-rtep-window-1-reliability-analysis-report.pdf.

[44] Indicated PJM Transmission Owners at 21-29.

[45] Exec. Order No. 14156, 90 Fed. Reg. 8433 (Jan. 20, 2025).

facilities.[46] Similar problems will arise in the large load context because, as MISO Transmission Owners observe, "large load owners may be sophisticated in other technical fields," but they "typically have not been involved in construction of transmission facilities and are not in the same position as transmission owners when it comes to knowledge and expertise regarding transmission facilities."[47] The generation context offers only a weak analogy because generation customers may have at least *some* relevant technical expertise, while large load customers do not.[48]

The NYTOs additionally discuss that an extended option to build will create confusion over the assignment of protection and control responsibilities, and may introduce unexpected costs for transmission owners in case of customer errors in design, building, and upkeep.[49] And it will create another inefficiency by having inexperienced large load customers navigate state regulation of siting and other retail obligations.[50] Finally, large load customers will also approach construction with no knowledge of RTO and transmission owner planning processes.

Some of these problems might be ameliorated if large load customers were required to adhere to NERC reliability standards. But such a requirement will still not remove the obligation of transmission owners to protect or maintain the reliability of the transmission system or obey NERC rules. So the ultimate task of ensuring compliance will again fall on transmission owners. Transmission owners can ensure compliance much more efficiently when they are the ones building facilities in the first place. It is not just or reasonable to make transmission owners responsible for reliability of network upgrades constructed by others.

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[46] PPL at 15.

[47] MISO Transmission Owners at 27.

[48] SPP at 9-10.

[49] NYTOs at 17-18.

[50] *Id.* at 18; SPP at 21 ("the Commission should be mindful of state laws that restrict who can build and own transmission").

If the Commission nonetheless believes that large load customers should have some option to build, the option should be limited to transmission interconnection facilities and stand-alone network upgrades, *not* energized facilities or any facility beyond the interconnection point, like integrated, networked transmission lines. National Grid plc ("National Grid") has adopted a similar approach, allowing large load customers to construct *only* interconnection facilities (using National Grid's applicable standards and approved contractor list), not network upgrades.[51]

Some commenters support granting large load customers an option to build, but their positions are unpersuasive. Advanced Energy United ("AEU") points to Order No. 845, arguing that an interconnection customer without an option to build may pay more or wait longer for the transmission provider to construct interconnection facilities and network upgrades.[52] AEU would also have the Commission extend the option to build to all stand-alone network upgrades, including those required for multiple interconnection customers, creating for large load customers a parallel rule to Order No. 2023-A (for generator interconnection customers).[53] The Solar Energy Industries Association likewise argues that the logic of Order No. 845 transfers directly to large load interconnection customers, while arguing that the Commission "should maintain the same protections for transmission providers, including the transmission provider's right to approve the engineering design, the equipment tests, and the construction of the network upgrades" as way of minimizing reliability concerns.[54] And Constellation Energy Generation, LLC ("Constellation") offers the same analogy to the generator option-to-build.[55] Finally, the Southeast Public Interest

[51] National Grid at 17.

[52] AEU at 25.

[53] *Id*. at 25-26.

[54] Solar Energy Industries Association at 20.

[55] Constellation at 25-26.

Organizations agree and especially hope for lower costs. They argue that large load customer building avoids any incentive for transmission owners to "gold-plate" by choosing higher-cost venders to inflate the rate base.[56]

But as explained above, the analogy to Order No. 845 only emphasizes the defects of a large load customer option to build. The generator option to build is largely a failed experiment, with delays, reliability issues, and consequently high costs. As explained above, large load customers have even less experience than generators do with NERC standards and other technical issues, and if they build, the transmission owner will still have to exercise oversight, resulting in unnecessary extra costs.

II. CONCLUSION

FirstEnergy appreciates the opportunity to submit reply comments on the ANOPR, and respectfully requests that the Commission act in accordance with these reply comments, as well as FirstEnergy's initial comments.

Respectfully submitted,

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[56] Southern Environmental Law Center, Appalachian Voices, North Carolina Sustainable Energy Association, South Carolina Coastal Conservation League, and Southern Alliance for Clean Energy, at 48-49.